

A Stick-Number Lower Bound for Ropelength

ChatGPT as prompted by Alex Klotz

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Abstract

We derive the lower bound

$$\text{Rop}(K) \geq \pi(\text{stick}(K) - 1)$$

for a knot type K , under the standard unit-radius convention for ropelength. The key geometric input is the polygonal-approximation theorem for thick knots: a curve of length ℓ and thickness τ has the same knot type as a polygon with at most $\ell/(\pi\tau) + 1$ edges. We explain the normalization, the infimum argument, the role of exact stick numbers, and the specialization to the torus-knot families used in the accompanying comparisons with Diao's crossing-number bound.

1 Definitions and normalization

Let $\gamma : S^1 \rightarrow \mathbb{R}^3$ be an embedded closed curve. Its thickness is denoted by $\text{Thi}(\gamma)$. For a $C^{1,1}$ curve this can be characterized as the largest radius of an embedded normal tube; equivalently, it is controlled by both the radius of curvature and the relevant doubly-critical self-distance. The ropelength of a particular representative is

$$\mathcal{R}(\gamma) := \frac{\text{Len}(\gamma)}{\text{Thi}(\gamma)},$$

and the ropelength of a knot type K is

$$\text{Rop}(K) := \inf_{\gamma \in K} \mathcal{R}(\gamma).$$

The formulas below use the unit-radius convention. Thus, after rescaling any representative, we may assume $\text{Thi}(\gamma) = 1$ and then

$$\mathcal{R}(\gamma) = \text{Len}(\gamma).$$

If ropelength is instead normalized using unit diameter, every numerical ropelength in this document is divided by 2.

2 The polygonal approximation input

The following is the edge-number estimate proved in the thickness theory of Litherland, Simon, Durumeric, and Rawdon. It is the precise bridge between ropelength and stick number used here.

Lemma 1 (Thick-curve polygonal approximation). *Let γ be an embedded closed curve with length ℓ and thickness $\tau > 0$. There is a polygonal knot P ambient isotopic to γ with*

$$N(P) \leq \frac{\ell}{\pi\tau} + 1$$

edges.

Geometric mechanism. Partition γ into consecutive subarcs whose arclengths are at most $\pi\tau$, and join consecutive partition points by straight chords. The thickness condition gives the local tube and restricted-bending estimates needed to ensure that each such subarc can be replaced by its chord without creating a self-intersection or changing the ambient isotopy class. The resulting inscribed polygon is therefore isotopic to γ .

If N is the number of subarcs, then the partition can be chosen with

$$N \leq \frac{\ell}{\pi\tau} + 1.$$

The polygon has N edges, which proves the claim. $\square$

Remark 1. The nontrivial part of the lemma is the isotopy assertion. It is not a consequence of arclength partitioning alone: an arbitrary curve can have a short subarc whose chord is topologically unsafe. The thickness hypothesis is what supplies the excluded-volume and bending control. Accordingly, the lemma should be cited as the polygonal-approximation theorem rather than presented as an elementary consequence of the definition of thickness.

3 Derivation of the stick-based ropelength bound

Let $s(K) = \text{stick}(K)$ denote the minimum number of edges in any polygonal representative of K . Applying the lemma to an arbitrary representative $\gamma \in K$ gives

$$s(K) \leq \frac{\text{Len}(\gamma)}{\pi \text{Thi}(\gamma)} + 1.$$

Rearranging,

$$\frac{\text{Len}(\gamma)}{\text{Thi}(\gamma)} \geq \pi(s(K) - 1).$$

Since this holds for every representative of K , taking the infimum yields the main result.

Theorem 1 (Stick-based ropelength lower bound). *For every nontrivial knot type K ,*

$$\boxed{\text{Rop}(K) \geq \pi(\text{stick}(K) - 1).}$$

More generally, if a rigorous lower bound $s_0 \leq \text{stick}(K)$ is known, then

$$\text{Rop}(K) \geq \pi(s_0 - 1).$$

Proof. The inequality in the preceding display holds for every representative $\gamma \in K$. Taking the infimum over γ proves the result. $\square$

Remark 2 (Exactness requirement). To use the theorem numerically, one needs an exact stick number or a rigorous lower bound for it. A polygonal construction with s edges proves only $\text{stick}(K) \leq s$ and therefore does not, by itself, imply $\text{Rop}(K) \geq \pi(s - 1)$. The latter inference is valid when the construction is accompanied by a proof that no polygon with fewer edges represents K .

4 Torus-knot specialization

Write $T(q, p)$ with $2 \leq p < q$ and $\gcd(p, q) = 1$. Its crossing number is

$$\text{Cr}(T(q, p)) = q(p - 1).$$

In the ranges where the torus-knot stick-number theorem gives

$$\text{stick}(T(q, p)) = \begin{cases} 2q, & p < q < 2p, \\ 4p, & 2p < q < 3p, \end{cases}$$

the theorem above gives

$$\text{Rop}(T(q, p)) \geq \begin{cases} \pi(2q - 1), & p < q < 2p, \\ \pi(4p - 1), & 2p < q < 3p. \end{cases}$$

Dividing by the three-quarter power of crossing number gives

$$\frac{\text{Rop}(T(q, p))}{\text{Cr}(T(q, p))^{3/4}} \geq \begin{cases} \frac{\pi(2q - 1)}{[q(p - 1)]^{3/4}}, & p < q < 2p, \\ \frac{\pi(4p - 1)}{[q(p - 1)]^{3/4}}, & 2p < q < 3p. \end{cases}$$

5 Worked example: $T(9, 4)$

Since $2p < q < 3p$ for $(q, p) = (9, 4)$, the exact stick number is

$$\text{stick}(T(9, 4)) = 4p = 16.$$

The crossing number and stick-based bound are therefore

$$\text{Cr}(T(9, 4)) = 9(4 - 1) = 27, \quad \text{Rop}(T(9, 4)) \geq \pi(16 - 1) = 15\pi.$$

Numerically,

$$15\pi \approx 47.12389, \quad \frac{15\pi}{27^{3/4}} \approx 3.97849.$$

For comparison, Diao's lower bound is

$$D(C) := \frac{17.334 + \sqrt{17.334^2 + 64\pi C}}{2},$$

so that

$$D(27) \approx 46.51254.$$

Thus, for this knot, the stick-derived bound is larger by approximately 0.61135.

6 Comparison principle

If another rigorous lower bound $B(K)$ is available, the strongest currently known bound from this collection is simply

$$\text{Rop}(K) \geq \max \left\{ \pi(\text{stick}(K) - 1), D(\text{Cr}(K)), B(K) \right\}.$$

In particular, using the quoted four-thirds-power estimate $B(C) = 2.135C^{3/4}$, the stick bound is strongest precisely when

$$\pi(\text{stick}(K) - 1) > \max \left\{ D(\text{Cr}(K)), 2.135 \text{Cr}(K)^{3/4} \right\}.$$

Comment from human: The coefficient 2.135 is greater than the standard published value of 1.105, the derivation can be found in another github document.

7 The family $T(P, 2)$

Assume $P \geq 3$ is odd, so that $T(P, 2)$ is a knot rather than a two-component torus link. Its crossing number is

$$\text{Cr}(T(P, 2)) = P.$$

Let $s_P := \text{stick}(T(P, 2))$. The stick-number theorem immediately gives the following general statement:

$$\boxed{\text{Rop}(T(P, 2)) \geq \pi(s_P - 1).}$$

Consequently, any rigorous lower bound $s_P \geq \sigma(P)$ implies

$$\text{Rop}(T(P, 2)) \geq \pi(\sigma(P) - 1).$$

7.1 The conditional $s_P = 2P$ calculation

Comment from human: This condition is wrong, alternating torus knots are known to have a lower stick number than $2P$, and section 7.2 has the real lower bound.

If one assumes the exact identity

$$s_P = 2P,$$

then substitution into the stick-based theorem gives

$$\boxed{\text{Rop}(T(P, 2)) \geq \pi(2P - 1).}$$

Since $\text{Cr}(T(P, 2)) = P$, the corresponding normalized lower bound is

$$\frac{\text{Rop}(T(P, 2))}{\text{Cr}(T(P, 2))^{3/4}} \geq \frac{\pi(2P - 1)}{P^{3/4}}.$$

As $P \rightarrow \infty$ this behaves as

$$\frac{\pi(2P - 1)}{P^{3/4}} = 2\pi P^{1/4} - \pi P^{-3/4},$$

which is not asymptotically a constant four-thirds-power coefficient.

7.2 Important status of the assumption

The identity $s_P = 2P$ must not be used as an unconditional theorem for large P without checking its precise hypotheses and parameter range. In particular, recent work constructs representatives of $T(P, 2)$ with at most

$$\left\lfloor \frac{2}{3}(P+1) \right\rfloor + 4$$

sticks for the tested values of P , and reports that these constructions match the observed stick number in that range. Such a construction is an upper bound on s_P and, whenever it is strictly below $2P$, rules out the claim $s_P = 2P$ for that value of P .

Therefore the logically valid alternatives are:

1. if a proof gives $s_P = 2P$, then the conditional formula $\text{Rop}(T(P, 2)) \geq \pi(2P - 1)$ is rigorous;
2. if only a polygon with m sticks is known, then one has $s_P \leq m$, which does *not* yield a ropelength lower bound;
3. if a rigorous lower bound $s_P \geq \sigma(P)$ is known, then the unconditional stick-based estimate is $\text{Rop}(T(P, 2)) \geq \pi(\sigma(P) - 1)$.

For comparison with Diao's bound, the latter is

$$D(P) = \frac{17.334 + \sqrt{17.334^2 + 64\pi P}}{2}.$$

Under the conditional assumption $s_P = 2P$, the stick estimate exceeds Diao's estimate whenever

$$\pi(2P - 1) > D(P).$$

This comparison is conditional on the exact stick-number identity; it does not follow from a construction with $2P$ sticks alone.

7.3 Comment from the human

If the stick number of alternating torus knots is asymptotically $2P/3$ then this stick-ropelength relation implies a lower bound of $L > (2\pi/3)C \approx 2.09C$. This is much stronger than Diao's lower bound of $C/56$ [5], but still much less than the (fairly tight) upper bound of $7.31C$ [6].

8 References

References

- [1] R. A. Litherland, J. Simon, O. Durumeric, and E. Rawdon, *Thickness of knots*, Topology and its Applications 91 (1999), 233–244. DOI: 10.1016/S0166-8641(97)00210-1.
- [2] Y. Diao, *The lower bounds of the lengths of thick knots*, Journal of Knot Theory and Its Ramifications 12 (2003), 1–16. DOI: 10.1142/S0218216503002275.
- [3] C. Adams, B. Brennan, D. Greilsheimer, and A. Woo, *Stick numbers and composition of knots and links*, Journal of Knot Theory and Its Ramifications 6 (1997), 149–161.

- [4] J. Cantarella, A. Rechnitzer, H. Schumacher, and C. Shonkwiler, *New upper bounds for stick numbers*, arXiv:2508.18263 (2025).
- [5] Y. Diao, *The ropelength conjecture of alternating knots*, Mathematical Proceedings of the Cambridge Philosophical Society, 177(2), 2024
- [6] H Kim, S Oh, Y Huh. *Efficiency of non-identical double helix patterns in minimizing ropelength of torus knots* Physica Scripta, 2024.